



Designing and Building AI Products and Services

Module 3 Office Hours
Bruce Lam



Agenda

01 NN 101

02 Tokenization &
Embedding

03 What's an LLM
Parameter?

04 Explainability vs.
Performance

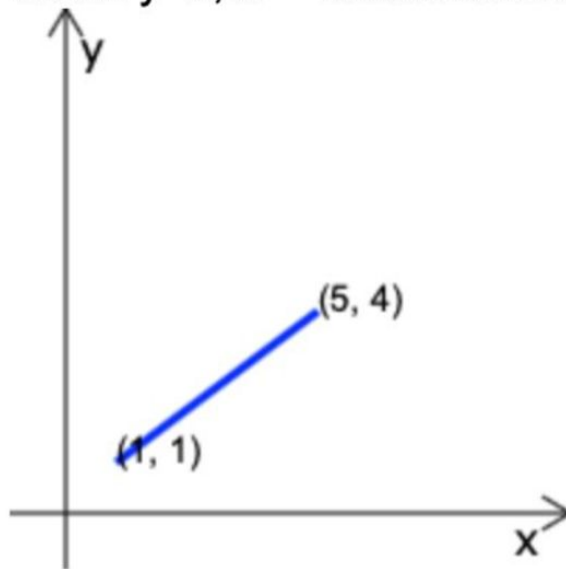
Regression

Equation of the line:

$$y = 0.75x + 0.25$$

When $x=0$, $y = 0.25$

When $y=0$, $x = -0.3333333333333333$



$$y = mx + b$$

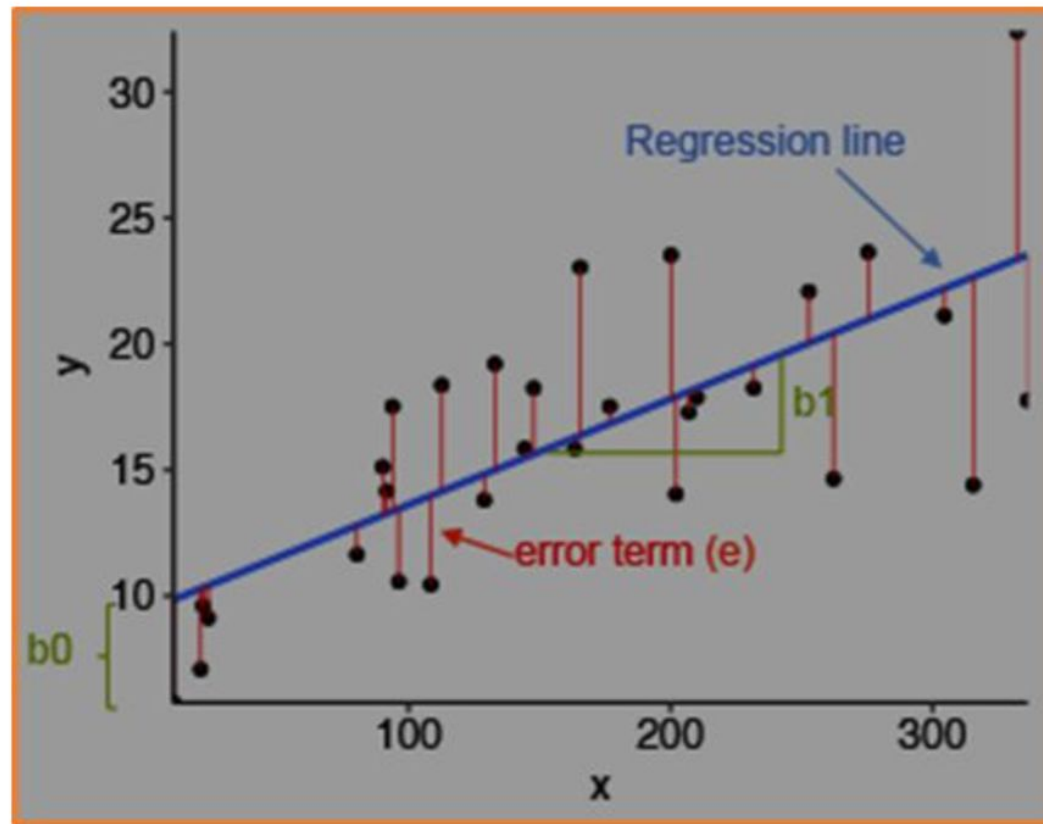
slope

y-intercept

<https://www.calculator.net/slope-calculator.html?type=2&x2=1&y2=1&d2=5&m2=0.75&th2=&x=57&y=27#onepoint>

[Slope-Intercept Form of a Straight Line \(\$y = mx + b\$ \) | ChiliMath](#)

Regression



Estimated (or predicted) y value

Estimate of the regression intercept

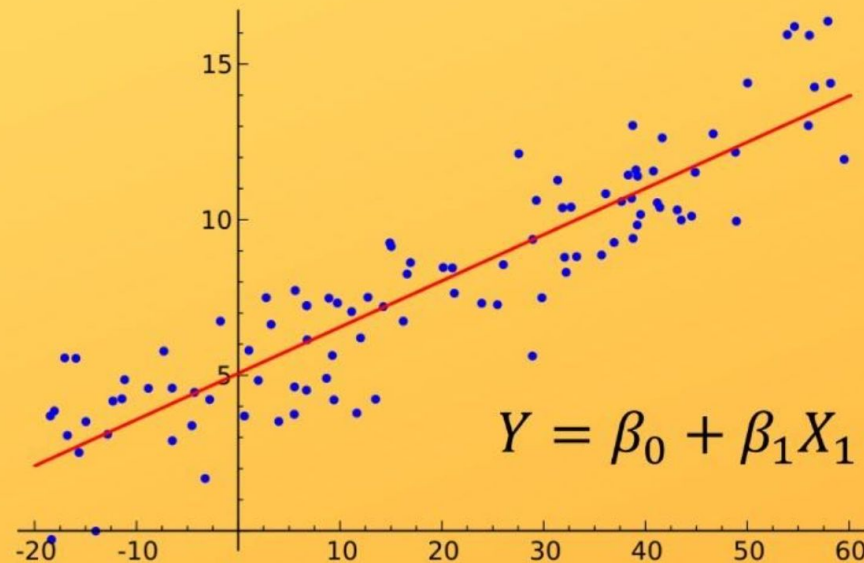
Estimate of the regression slope

Independent variable

Error term

$$y_i = b_0 + b_1 x + e$$

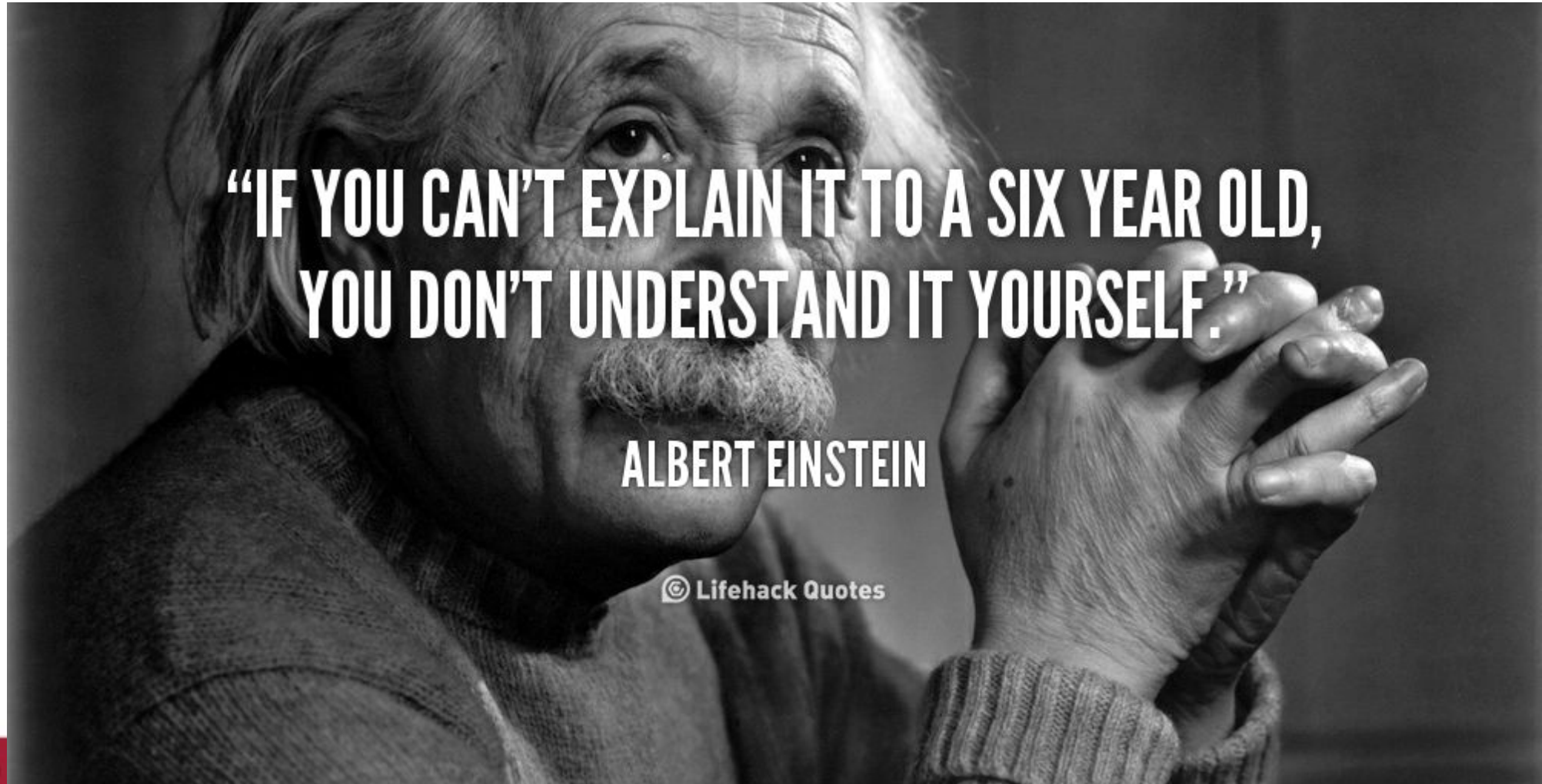
Multiple Linear Regression



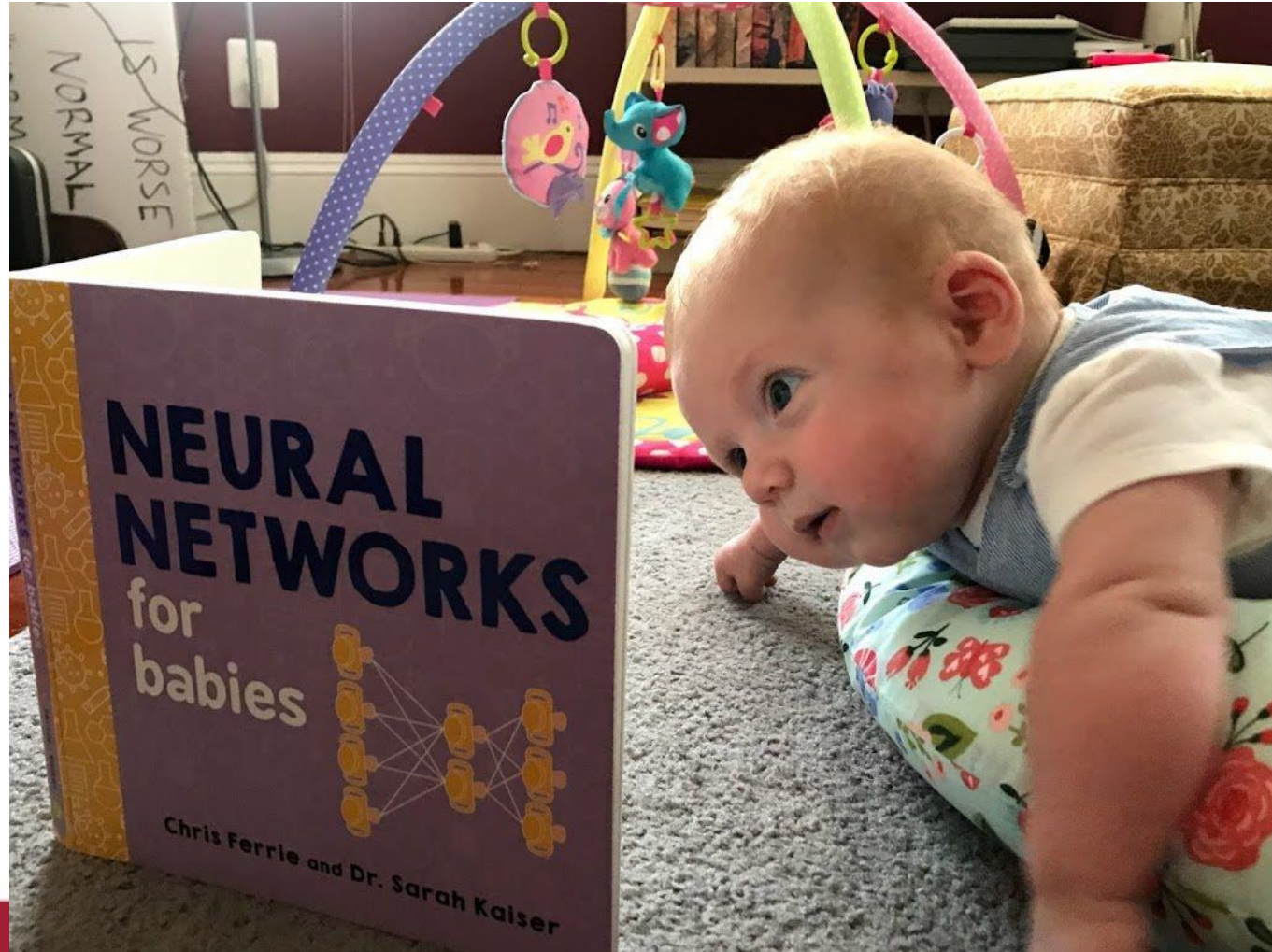
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

number of predictors

Neural Networks



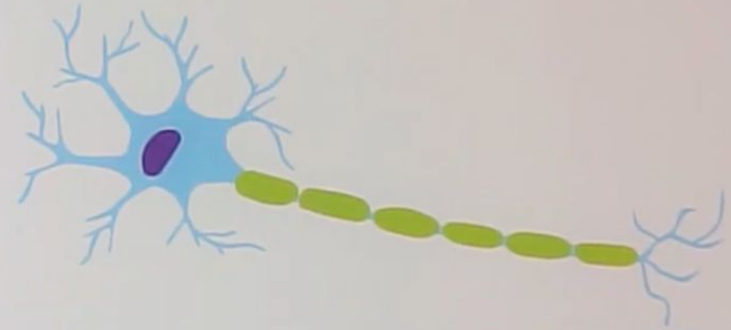
Neural Networks for Babies



Neural Networks for Babies

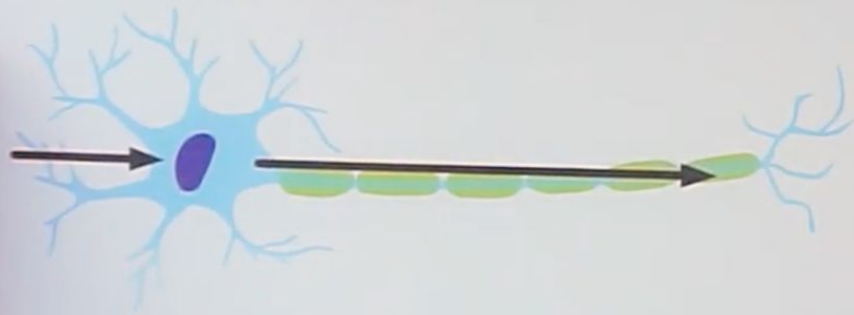


This is a ball.

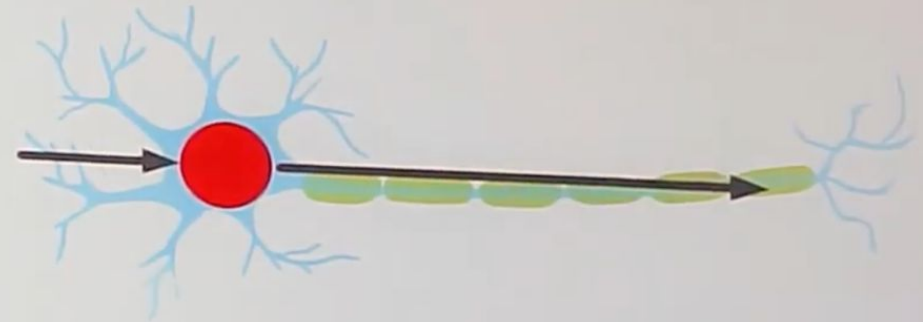


**This is a neuron. It sends
messages throughout your body.**

Neural Networks for Babies



Give the neuron input and output
and it can help us learn!



Give the ball input and output
and it acts like a neuron.

Neural Networks for Babies



The neuron can have one input and output,

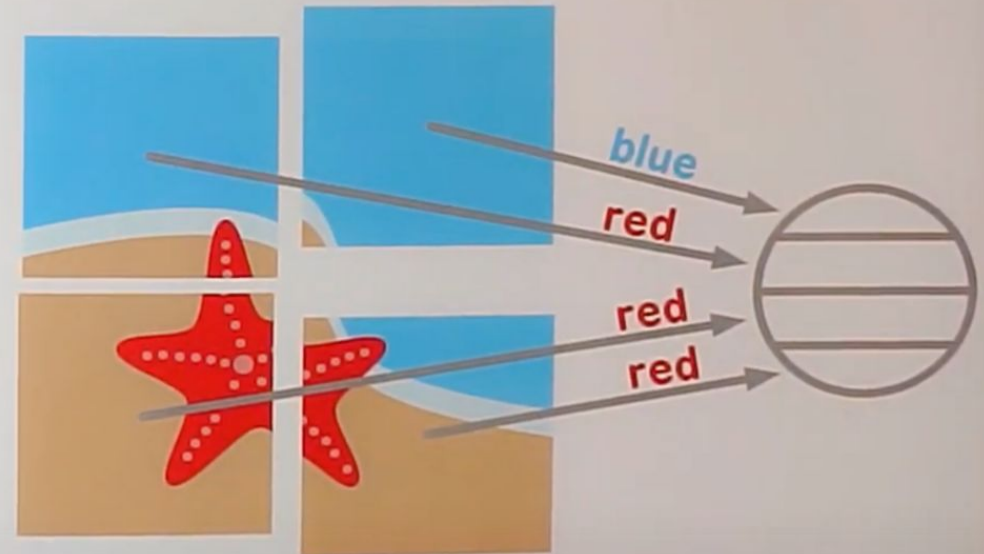


or many!

Neural Networks for Babies

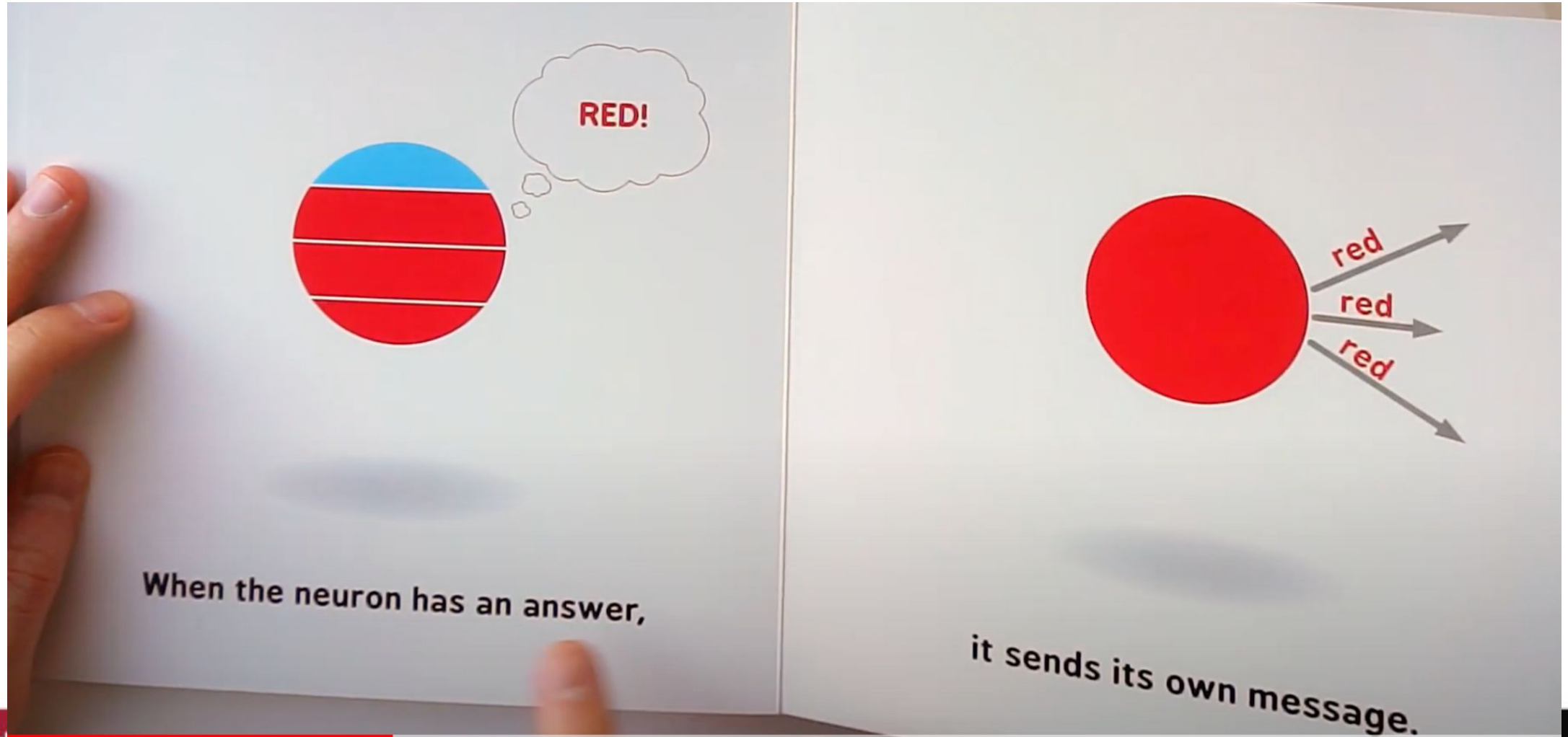


Is there a red animal in this picture?



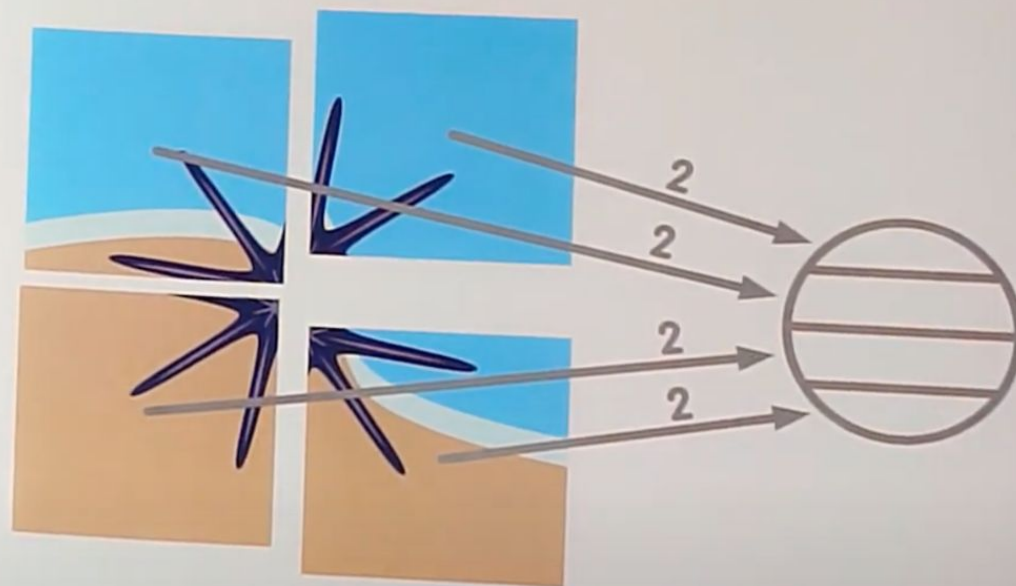
The neuron can decide based on its input.

Neural Networks for Babies



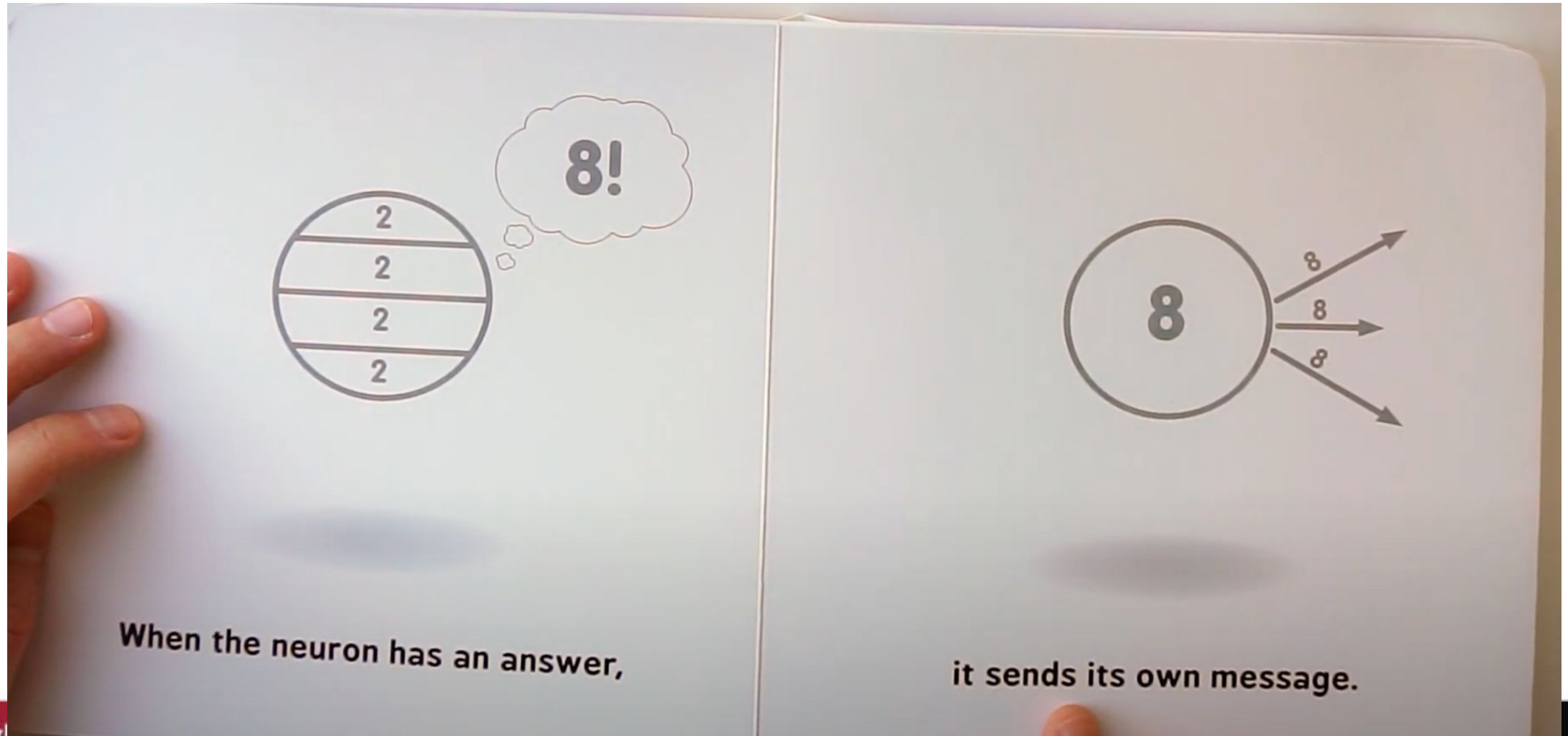


Does this animal have 8 arms?

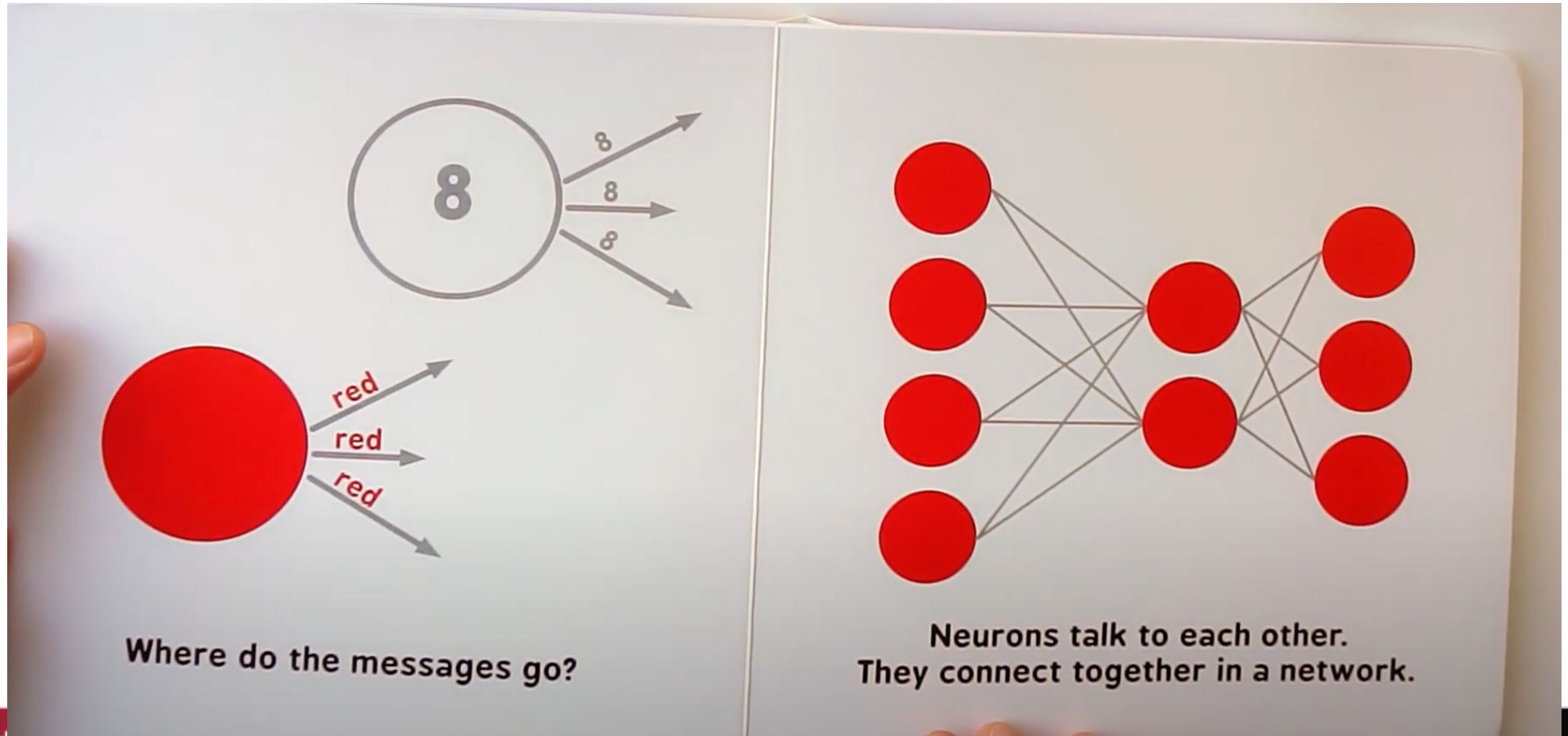


The neuron can decide based on its input.

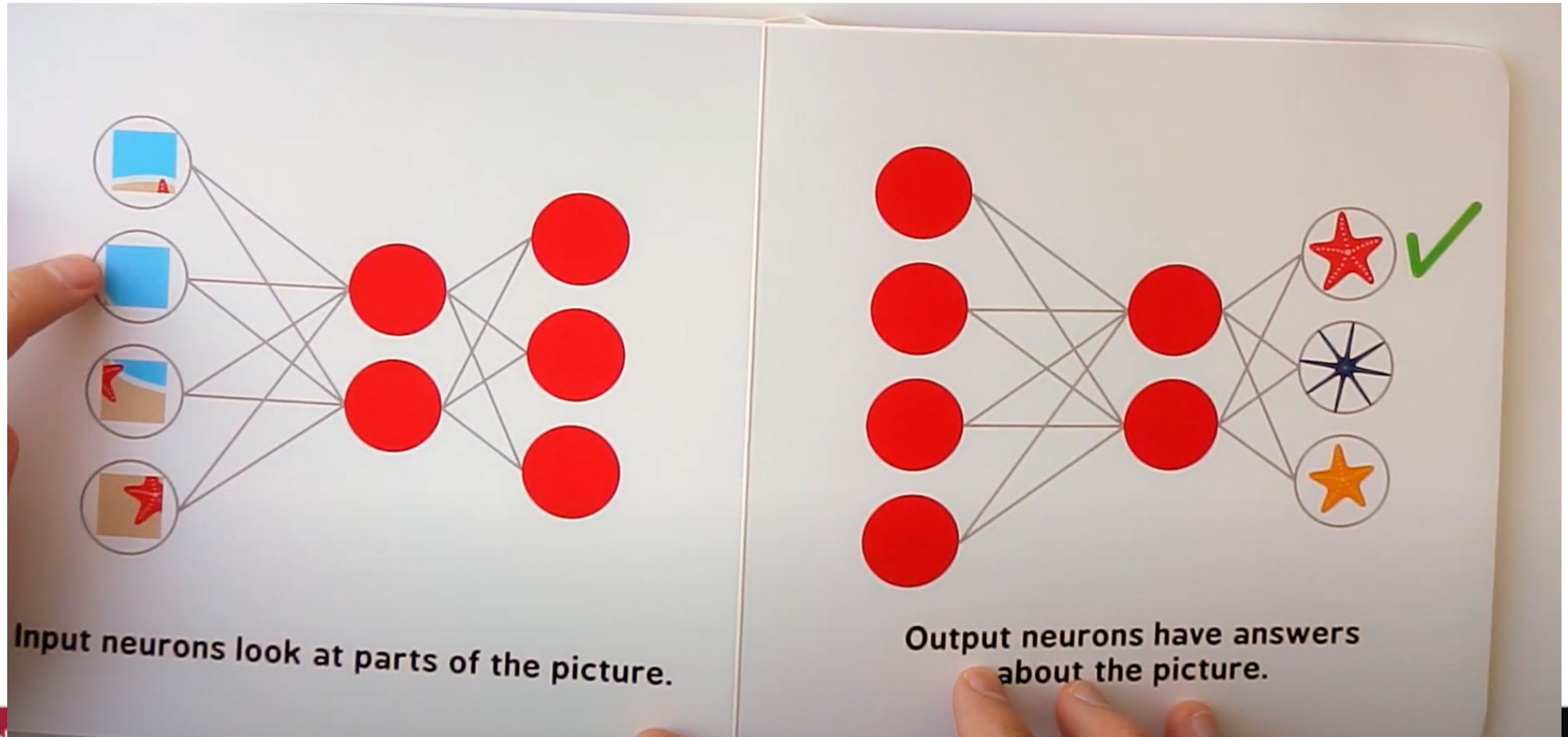
Neural Networks for Babies

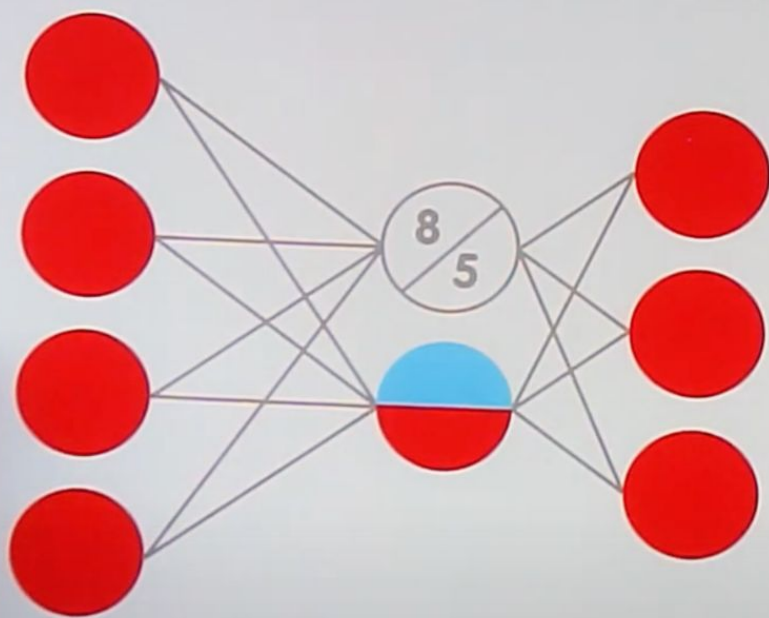


Neural Networks for Babies

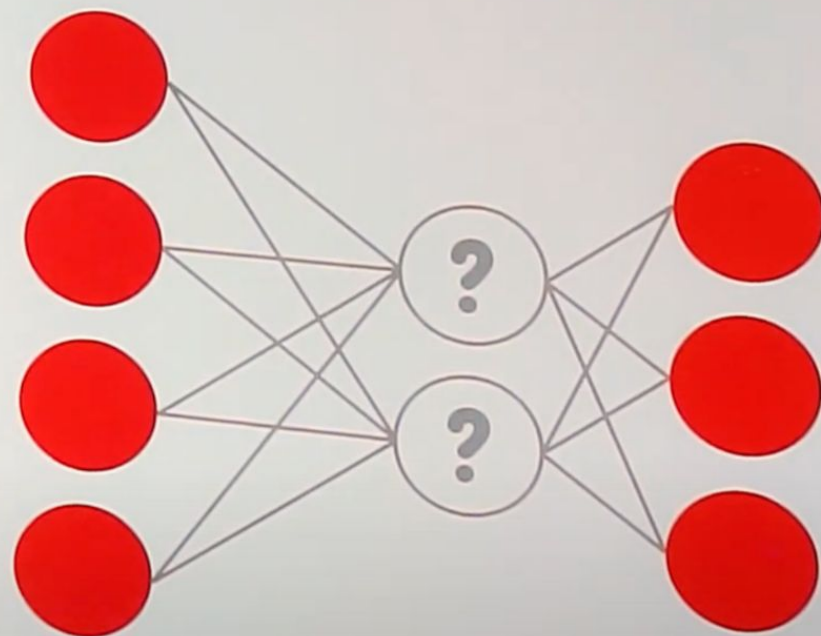


Neural Networks for Babies



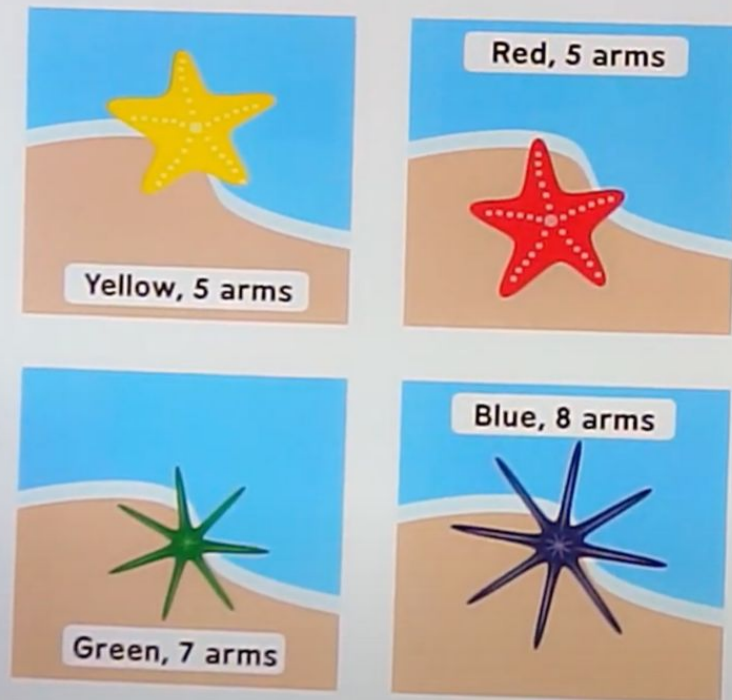


Neurons in between don't see the pictures or give answers; they are hidden.

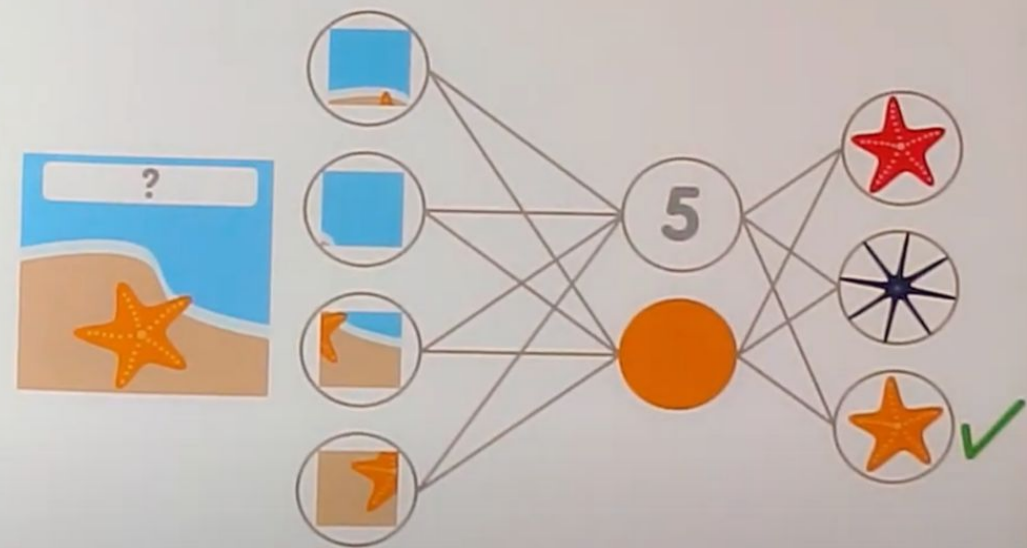


How do the hidden neurons learn to decide?

Neural Networks for Babies

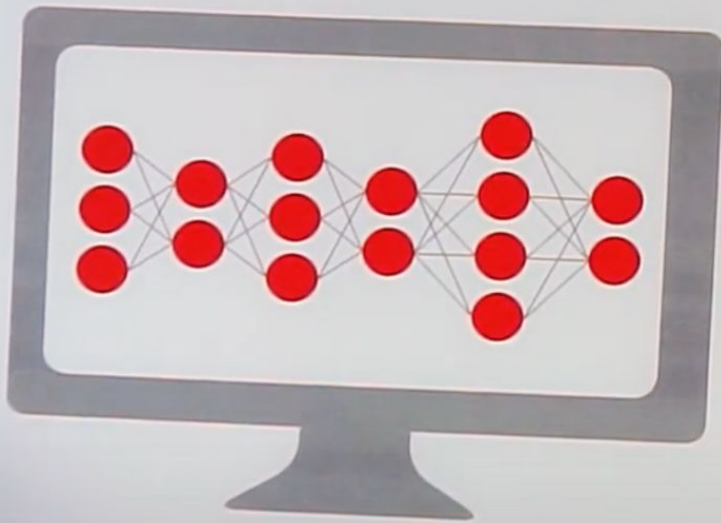


Training data can have correct labels on them.



After training, the network has learned to label new pictures.

Neural Networks for Babies



A really big network can solve even harder problems with the help of computers!

Now you know
**NEURAL
NETWORKS!**

Vectorization/Tokenization

GENERATIVE AI —

A jargon-free explanation of how AI large language models work

Want to really understand large language models? Here's a gentle primer.

TIMOTHY B. LEE AND SEAN TROTT - 7/31/2023, 4:00 AM



<https://arstechnica.com/science/2023/07/a-jargon-free-explanation-of-how-ai-large-language-models-work/>

<https://platform.openai.com/tokenizer>

Word vectors

To understand how language models work, you first need to understand how they represent words. Humans represent English words with a sequence of letters, like C-A-T for "cat." Language models use a long list of numbers called a "word vector." For example, **here's one way** to represent cat as a vector:

[0.0074, 0.0030, -0.0105, 0.0742, 0.0765, -0.0011, 0.0265, 0.0106, 0.0191, 0.0038, -0.0468, -0.0212, 0.0091, 0.0030, -0.0563, -0.0396, -0.0998, -0.0796, ..., 0.0002]

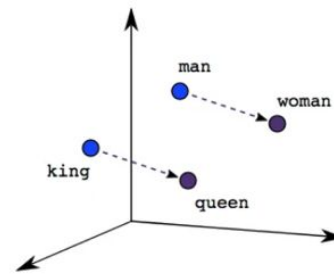
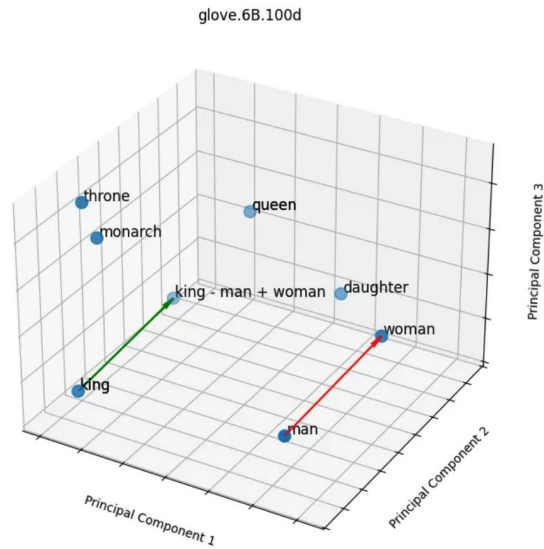
(The full vector is 300 numbers long—to see it all, **click here** and then click "show the raw vector.")

Why use such a baroque notation? Here's an analogy. Washington, DC, is located at 38.9 degrees north and 77 degrees west. We can represent this using a vector notation:

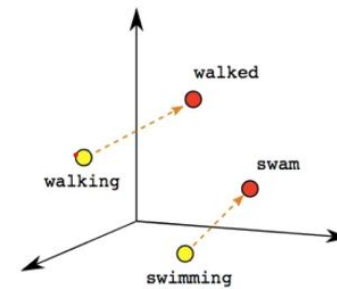
- Washington, DC, is at [38.9, 77]
- New York is at [40.7, 74]
- London is at [51.5, 0.1]
- Paris is at [48.9, -2.4]

This is useful for reasoning about spatial relationships. You can tell New York is close to Washington, DC, because 38.9 is close to 40.7 and 77 is close to 74. By the same token, Paris is close to London. But Paris is far from Washington, DC.

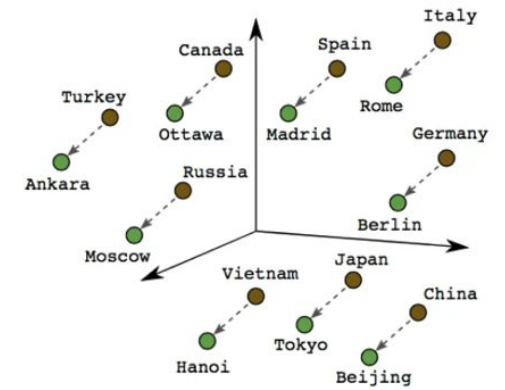
Embeddings



Male-Female



Verb Tense



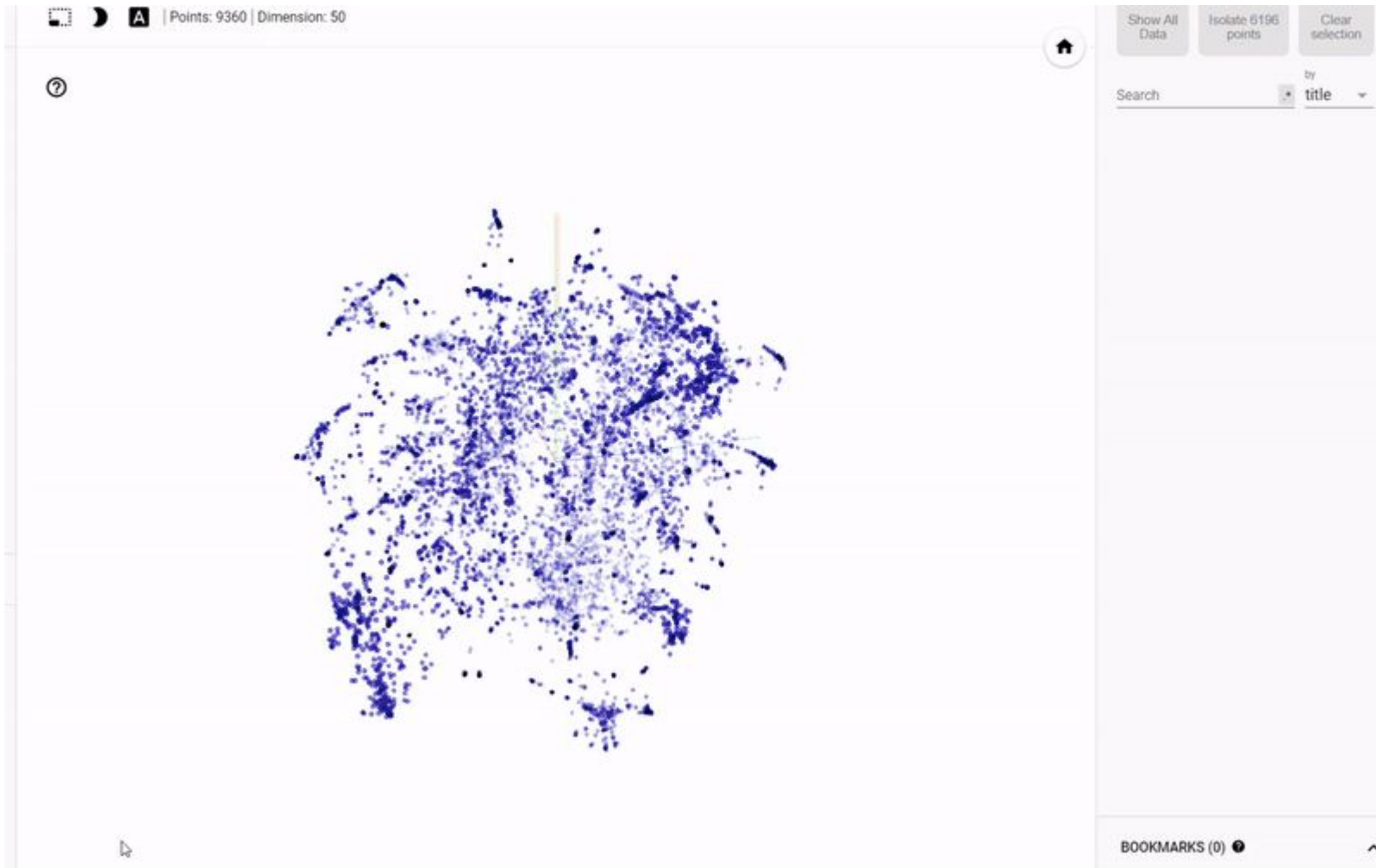
Country-Capital

GloVe embeddings 3D visualization. The embedding of king - man + woman is actually closer to the embedding of the word "king" than the word "queen".

https://medium.com/@mc_cheng/king-man-woman-is-not-queen-9cbf0afe6c72

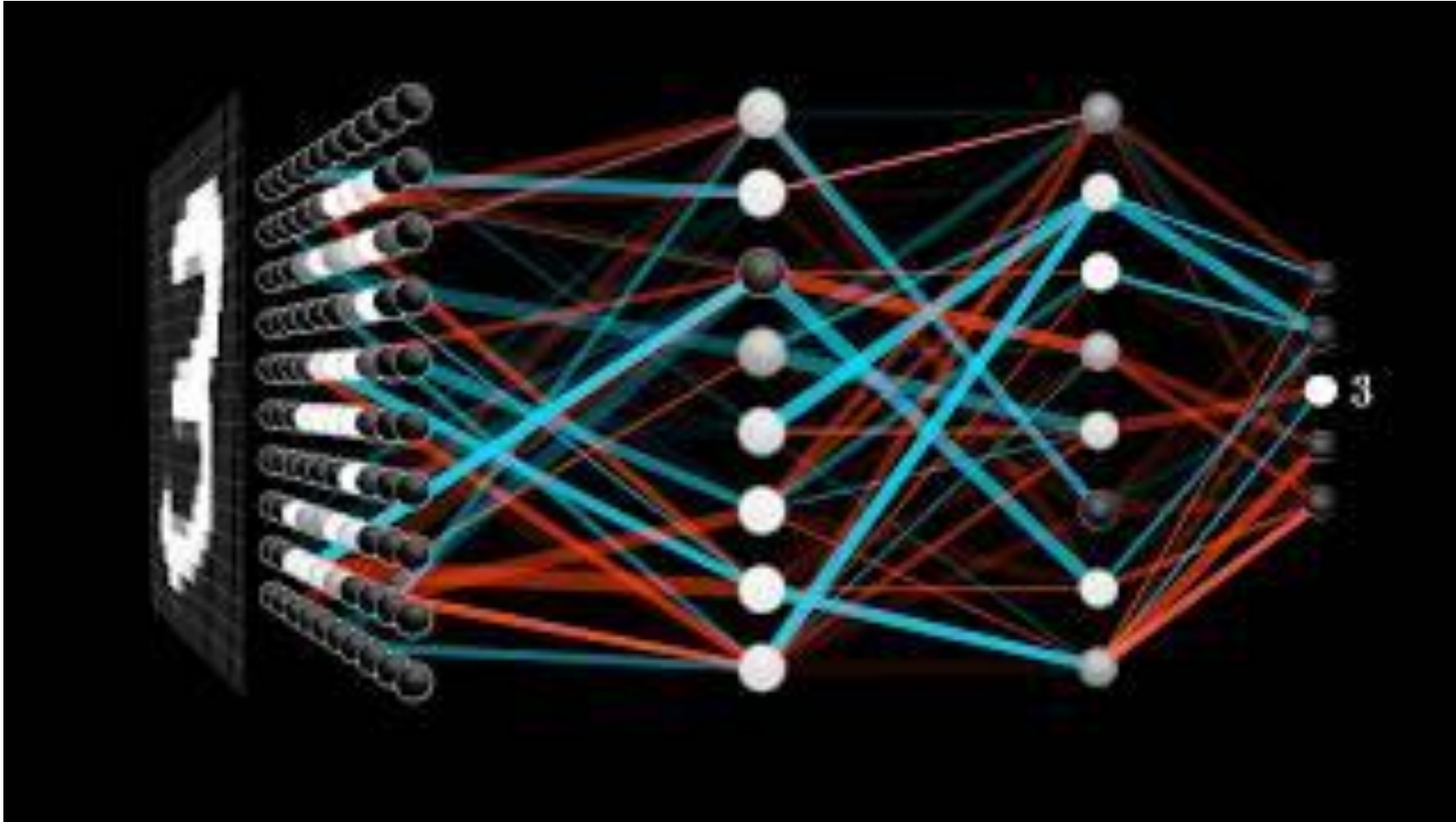
<https://www.linkedin.com/pulse/unpacking-word-embeddings-journey-through-modern-nlp-techniques-dczic/>

Embeddings



<https://medium.com/data-science/neural-network-embeddings-explained-4d028e6f0526>

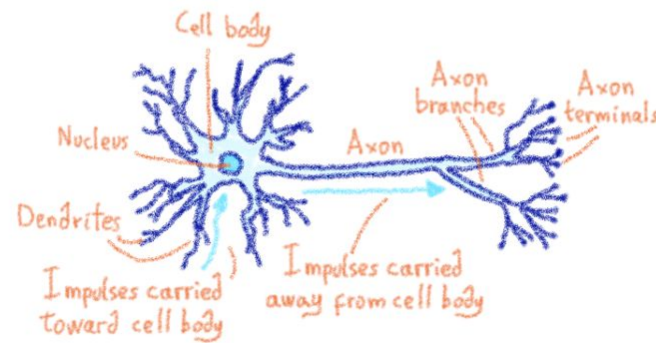
Neural Network Intuition



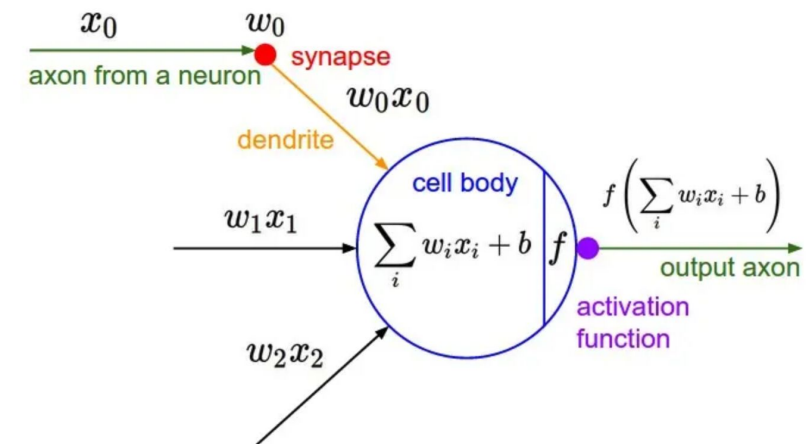
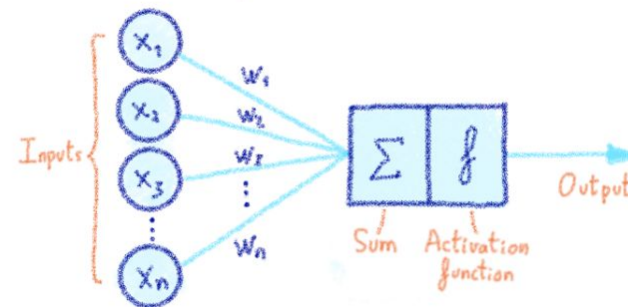
<https://www.youtube.com/watch?v=aircAruvnKk&list=WL&index=34&t=946s>

Perceptron (binary classification)

Biological Neuron



Artificial Neuron



https://cs231n.github.io/assets/nn1/neuron_model.jpeg

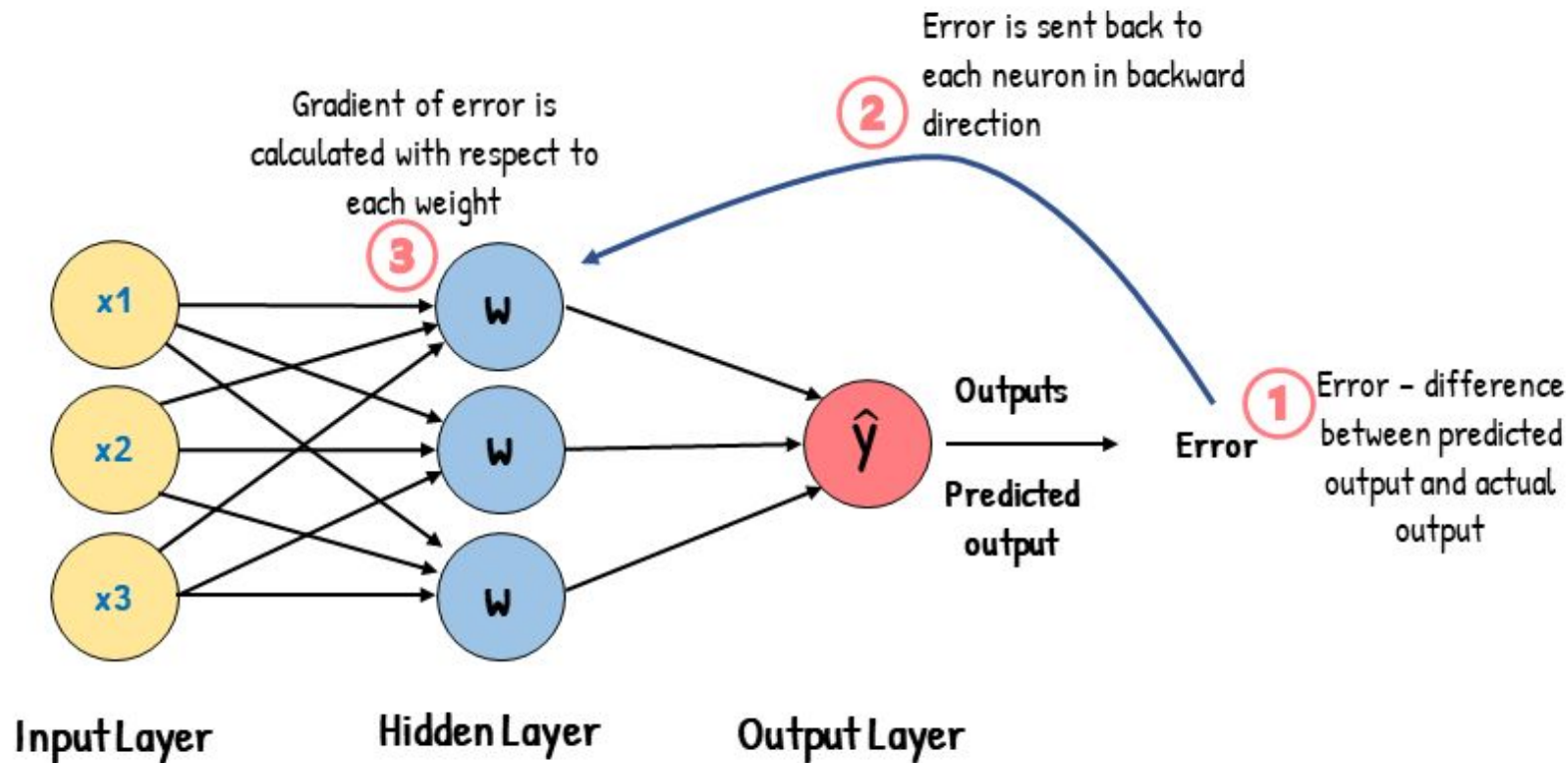


The Nobel Recognition

- **The Award:** Hinton shared the 2024 Nobel Prize in Physics with John J. Hopfield.
- **The Breakthrough:** While backpropagation allows deep neural networks to "learn" by adjusting internal weights based on errors, Hinton also played a pivotal role in inventing the **Boltzmann machine** and establishing **deep learning**. [NobelPrize.org +2](#)

Backpropagation

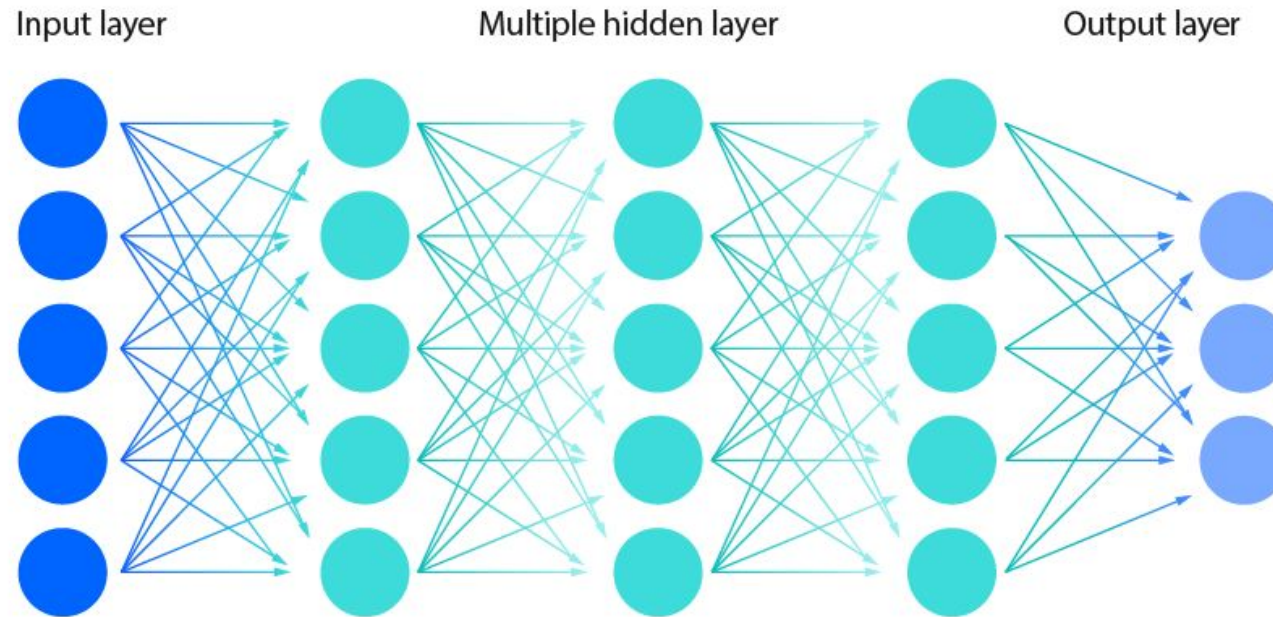
Backpropagation



<https://www.analyticsvidhya.com/blog/2023/01/gradient-descent-vs-backpropagation-whats-the-difference/>

Deep Learning

Deep neural network

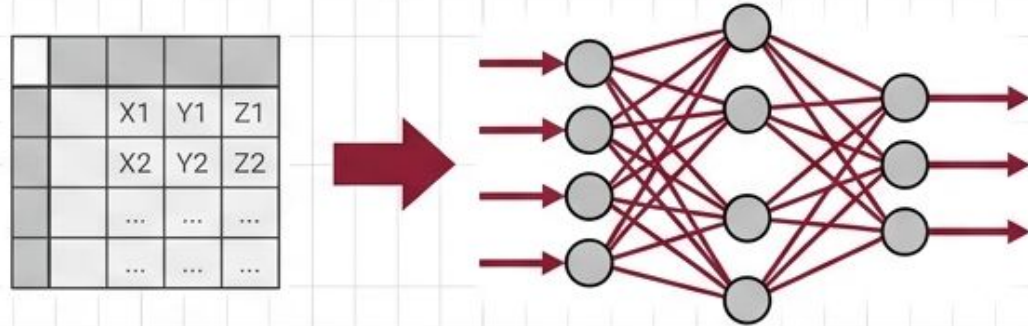


<https://towardsdatascience.com/deep-learning-illustrated-part-4-recurrent-neural-networks-d0121f27bc74>

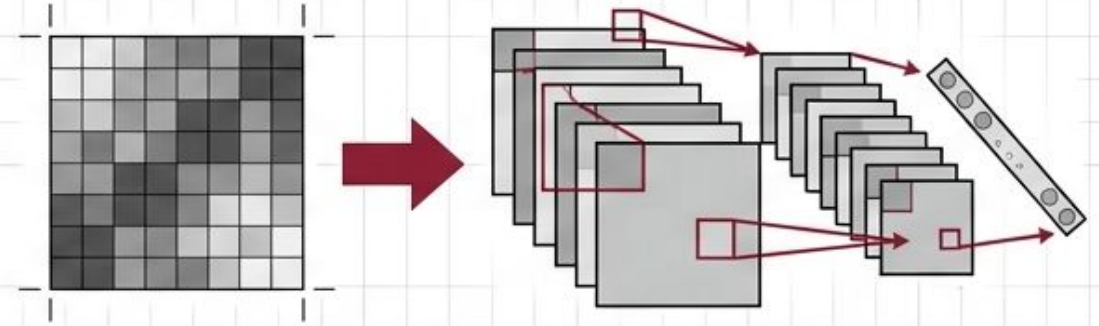
[What are Neural Networks? | IBM](#)

The Form of the Data Dictates the Architecture of the Network

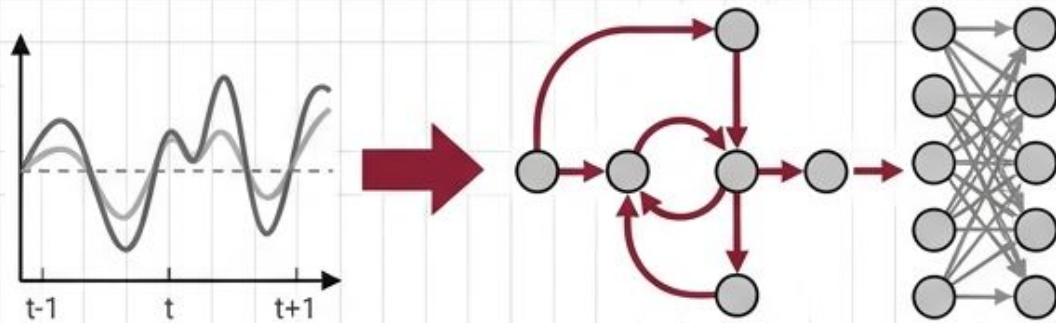
Neural networks are not generic brains; they are specialized structural filters.
Designing an AI system begins by **identifying the underlying topology of your data**.



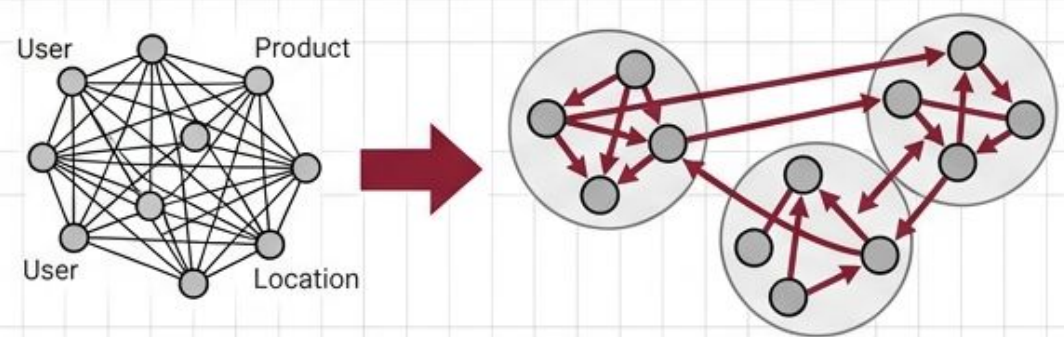
Tabular Data -> Feed-Forward Networks



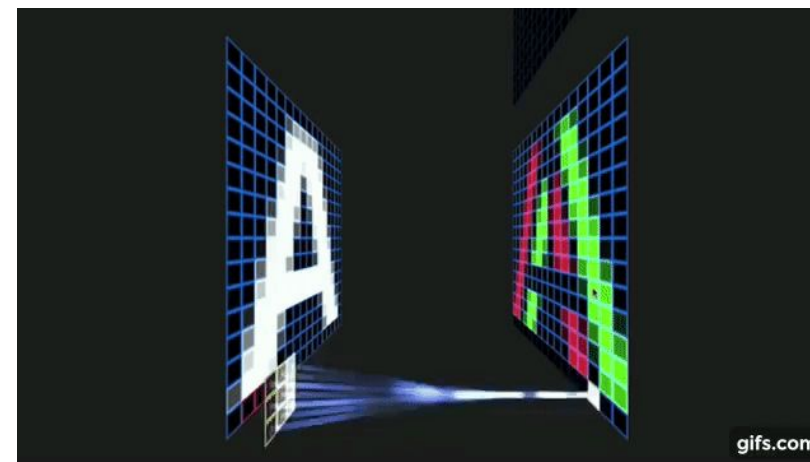
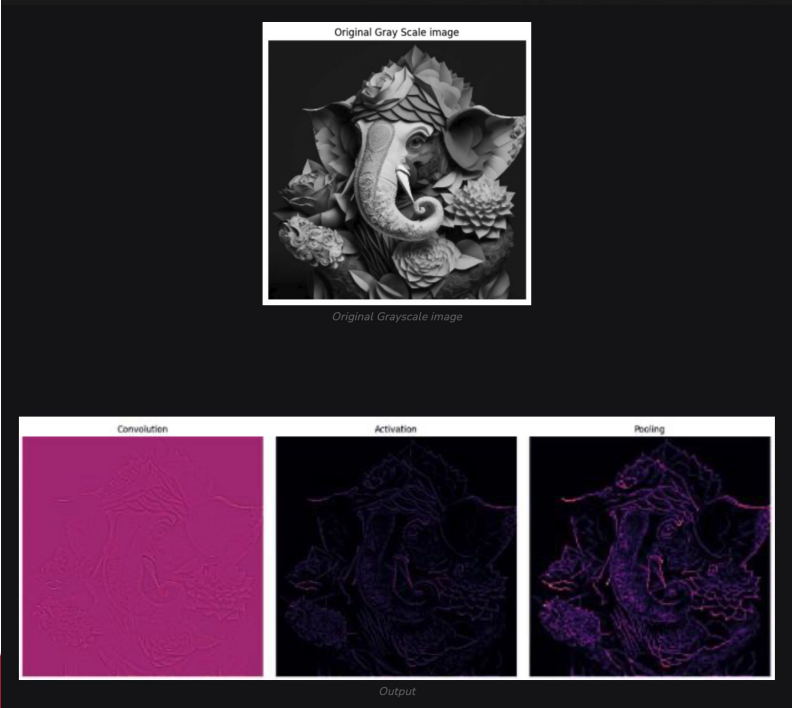
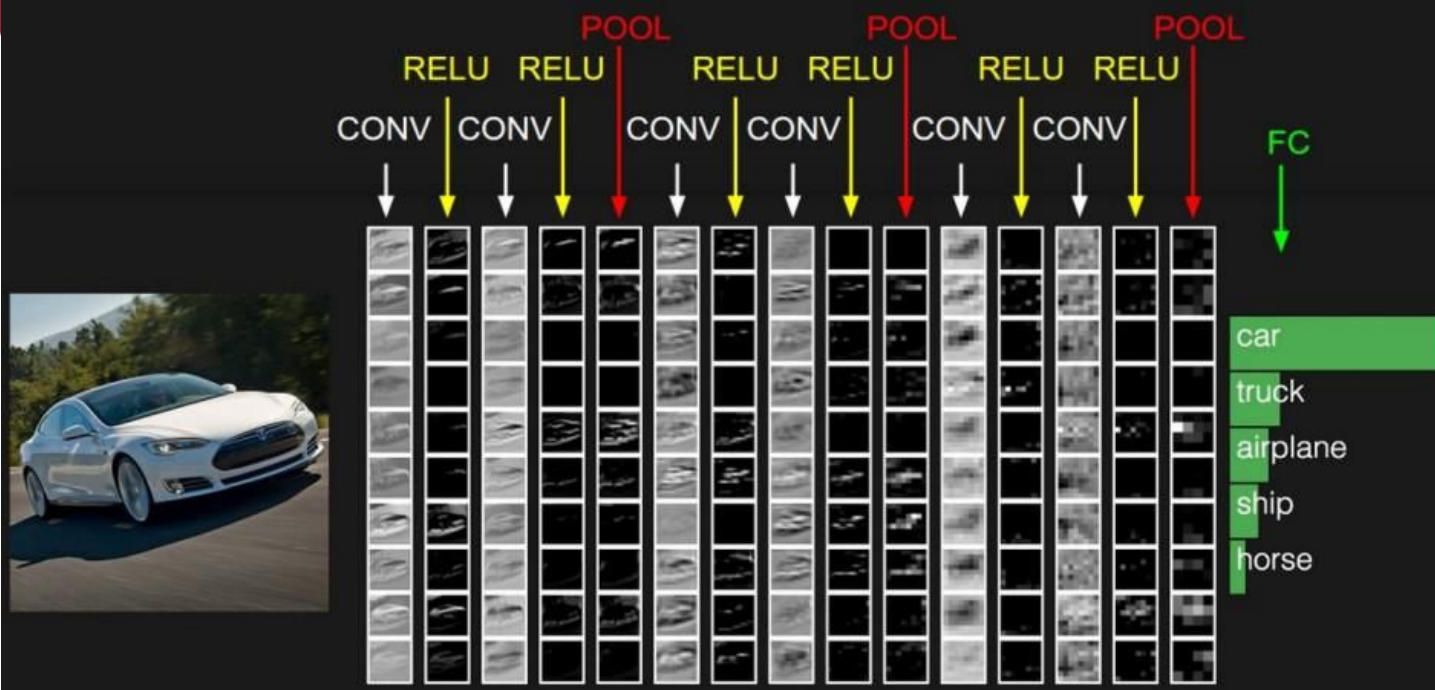
Grid / Spatial Data -> Convolutional Networks



Time / Sequential Data -> Recurrent Networks & Transformers



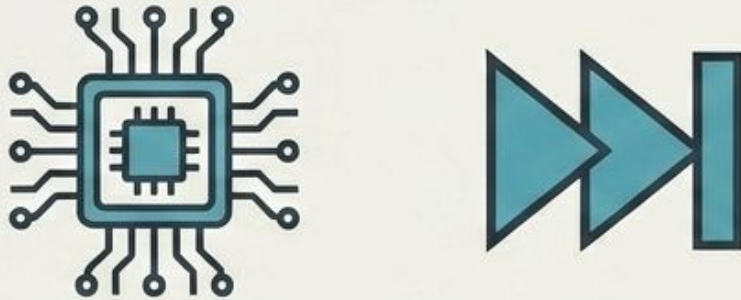
Graph / Relational Data -> Graph Neural Networks



<https://medium.com/brain-bites/introduction-to-convolutional-neural-networks-43b2ace7bc73>

Operational Trade-Offs: CNN vs. RNN

CNN (Space)



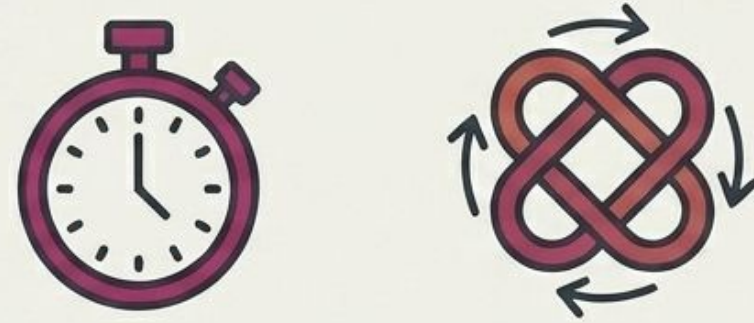
Pros:

- Exceptional feature extraction.
- Highly parallelizable (fast training on GPUs).

Cons:

- Demands immense computational power and high-quality data.
- Cannot inherently understand the order of inputs.

RNN (Time)



Pros:

- Retains sequential memory.
- Can handle variable-length inputs (One-to-Many, Many-to-Many).

Cons:

- Inherently sequential operations cannot be fully parallelized (slow training).
- Susceptible to vanishing/exploding gradients.



Tom Yeh (He/Him) · 1st
CS Prof | AI by Hand 🍌 | CU Boulder
Boulder, Colorado, United States · [Contact info](#)
70,415 followers · [500+ connections](#)
Gopal Renganathan, Mark Mims, and 3 other mutual connections

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Evolution of Deep Learning
by Geoff Hinton
2024 Nobel Prize Winner

Hand drawing
by Tom Yeh

Resources

AI & Machine Learning

Understanding neural networks with TensorFlow Playground

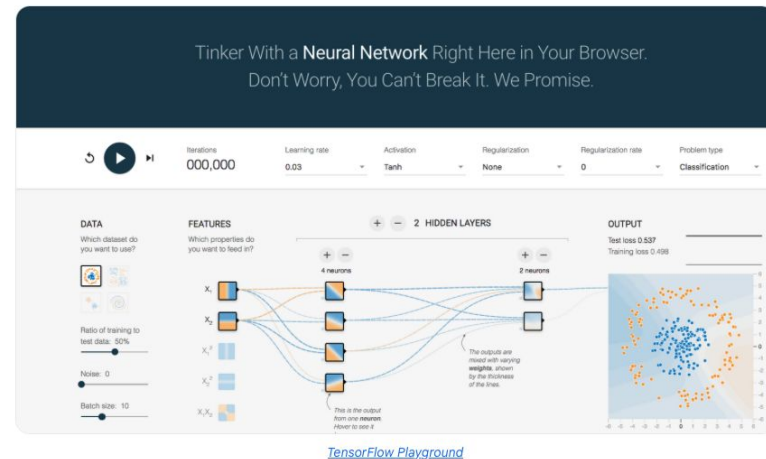
July 26, 2016

Kaz Sato

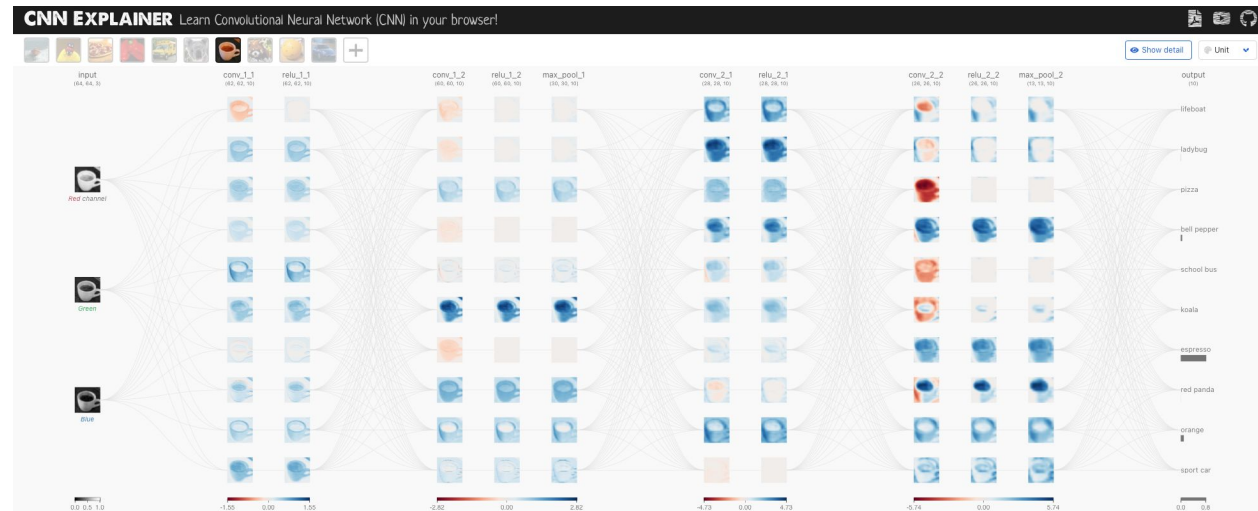
Developer Advocate, Cloud AI

You may have heard the buzz about neural networks and deep learning, and want to learn more. But when you learn about the technology from a textbook, many people find themselves overwhelmed by mathematical models and formulas. I certainly was.

For people like me, there's an awesome tool to help you grasp the idea of neural networks without any hard math: [TensorFlow Playground](https://tfplayground.tensorflow.org/), a web app written in JavaScript that lets you play with a real neural network running in your browser and click buttons and tweak parameters to see how it works.



<https://cloud.google.com/blog/products/ai-machine-learning/understanding-neural-network-s-with-tensorflow-playground>



What is a Convolutional Neural Network?

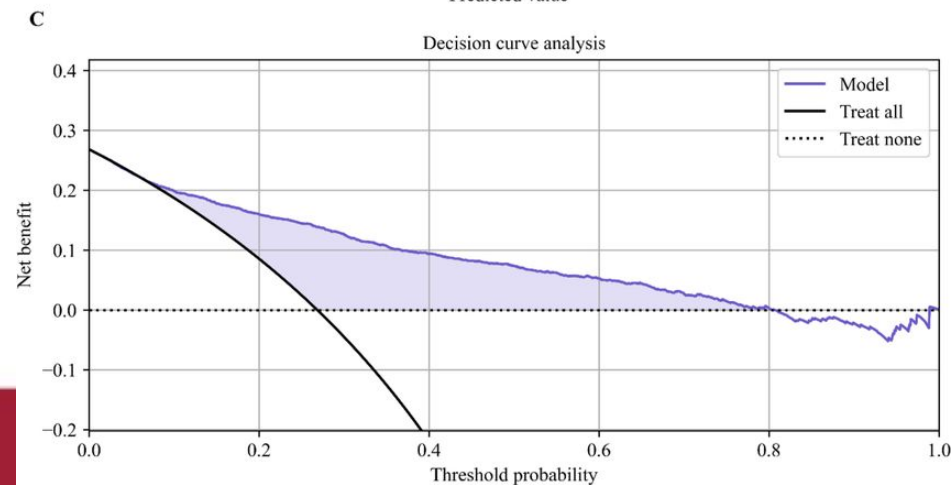
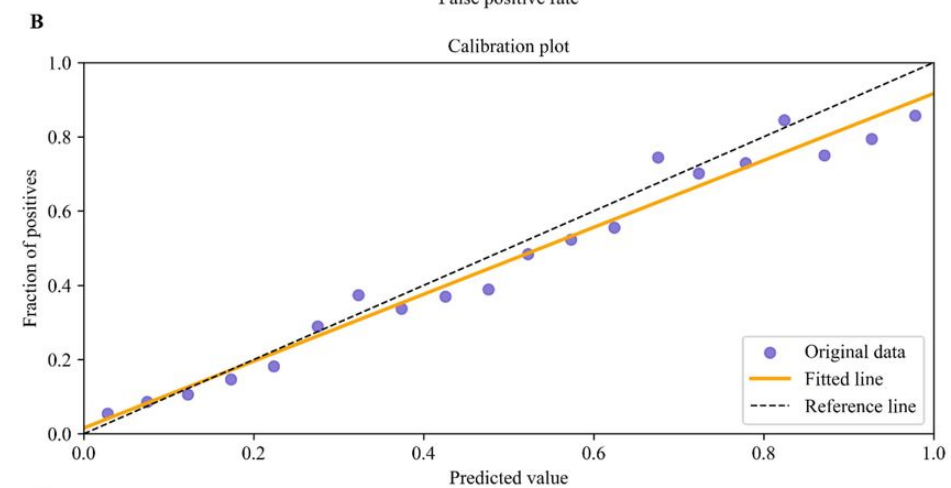
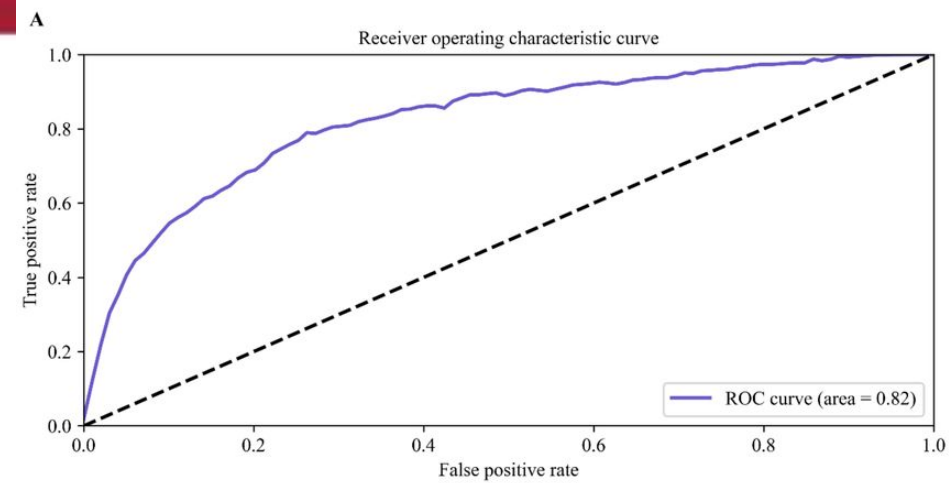
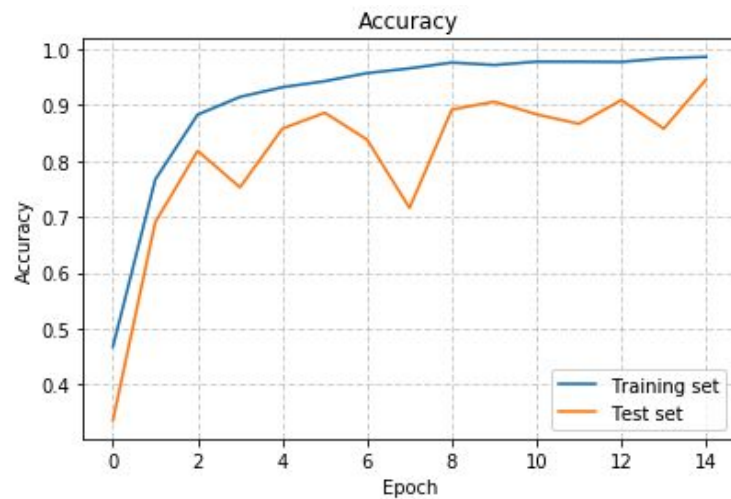
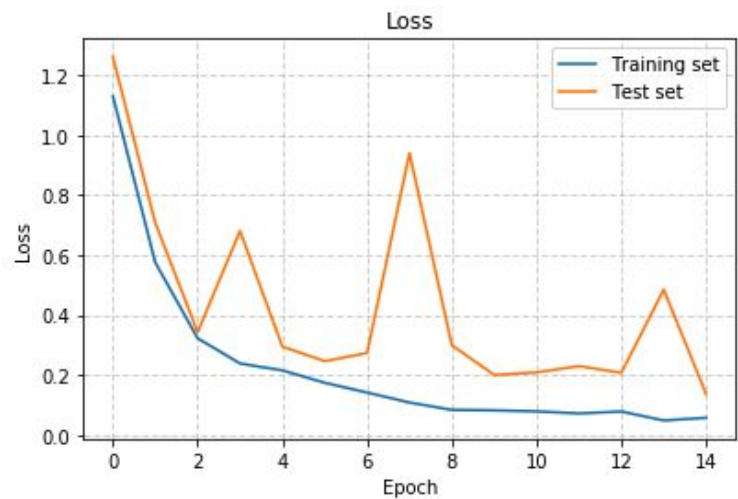
In machine learning, a classifier assigns a class label to a data point. For example, an *image classifier* produces a class label (e.g. bird, plane) for what objects exist within an image. A *convolutional neural network*, or CNN for short, is a type of classifier, which excels at solving this problem!

A CNN is a neural network: an algorithm used to recognize patterns in data. Neural Networks in general are composed of a collection of neurons that are organized in layers, each with their own learnable weights and biases. Let's break down a CNN into its basic building blocks.

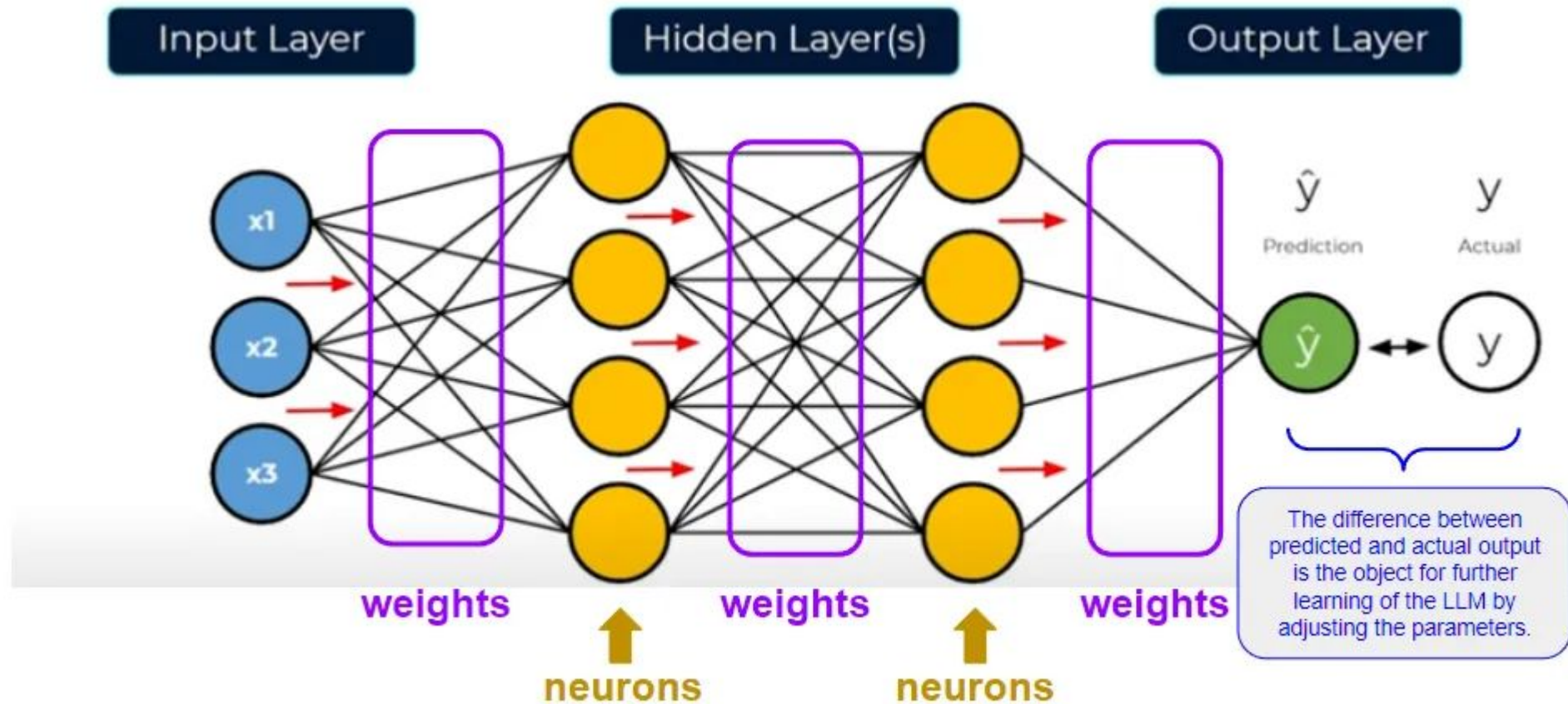
1. A **tensor** can be thought of as an n-dimensional matrix. In the CNN above, tensors will be 3-dimensional with the exception of the output layer.
2. A **neuron** can be thought of as a function that takes in multiple inputs and yields a single output. The outputs of neurons are represented above as the red → blue **activation maps**.
3. A **layer** is simply a collection of neurons with the same operation, including the same hyperparameters.
4. **Kernel weights and biases**, while unique to each neuron, are tuned during the training phase,

<https://poloclub.github.io/cnn-explainer/#:~:text=An%20interactive%20visualization%20system,about%20Convolutional%20Neural%20networks>

NN Performance metrics



What is an LLM Parameter?



https://lukaszostrowski.substack.com/p/parameters-in-an-llm-lets-shed-some?utm_medium=email

NN Learning Rate

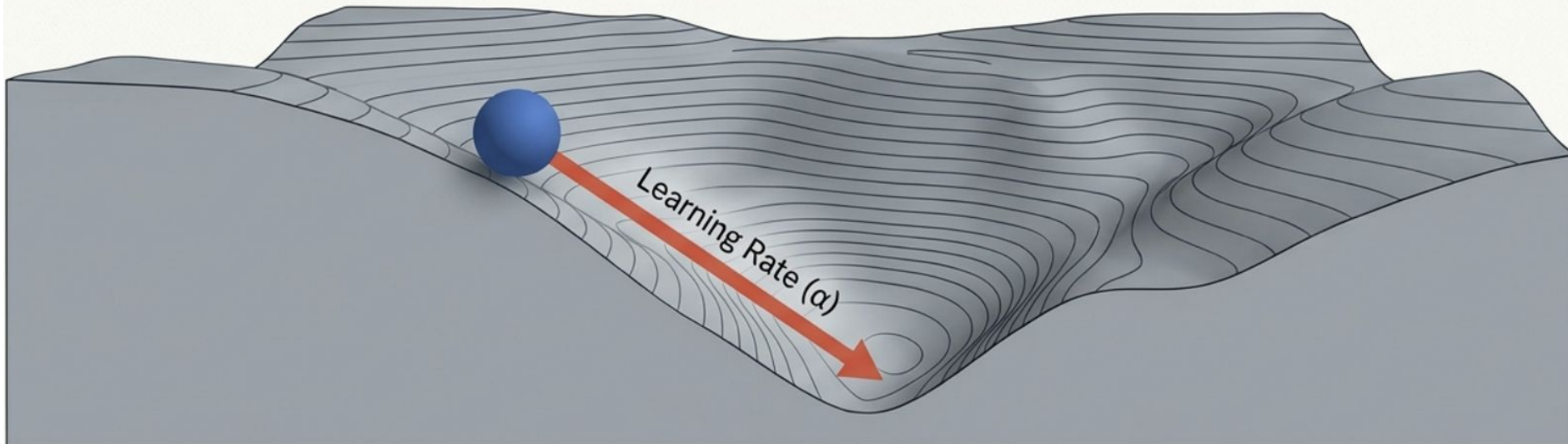
The Learning Rate Determines the Stride of Your Descent

The learning rate is the hyperparameter that controls the step size a neural network takes as it minimizes error during training.

Dictates how aggressively weights are updated.

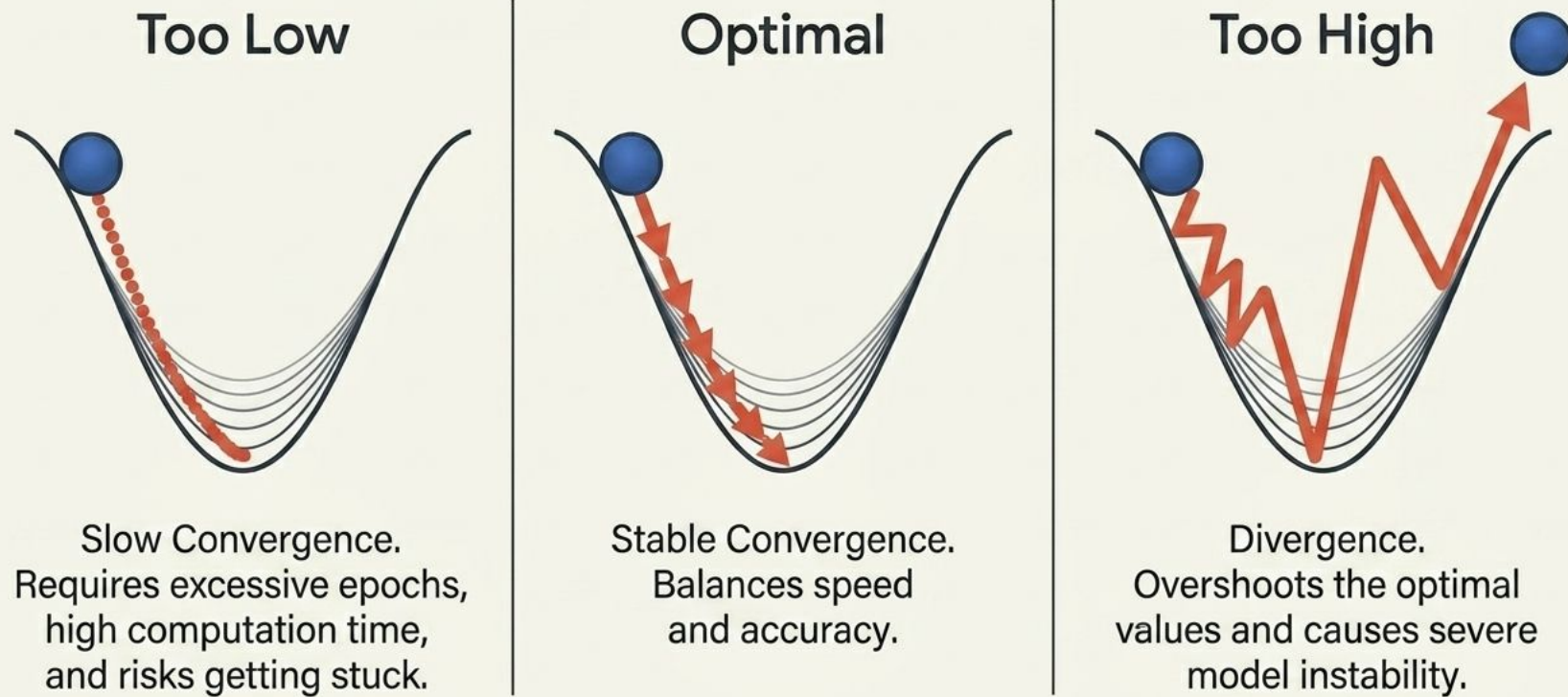
Directly affects the speed and stability of training.

Learning Rate (α)



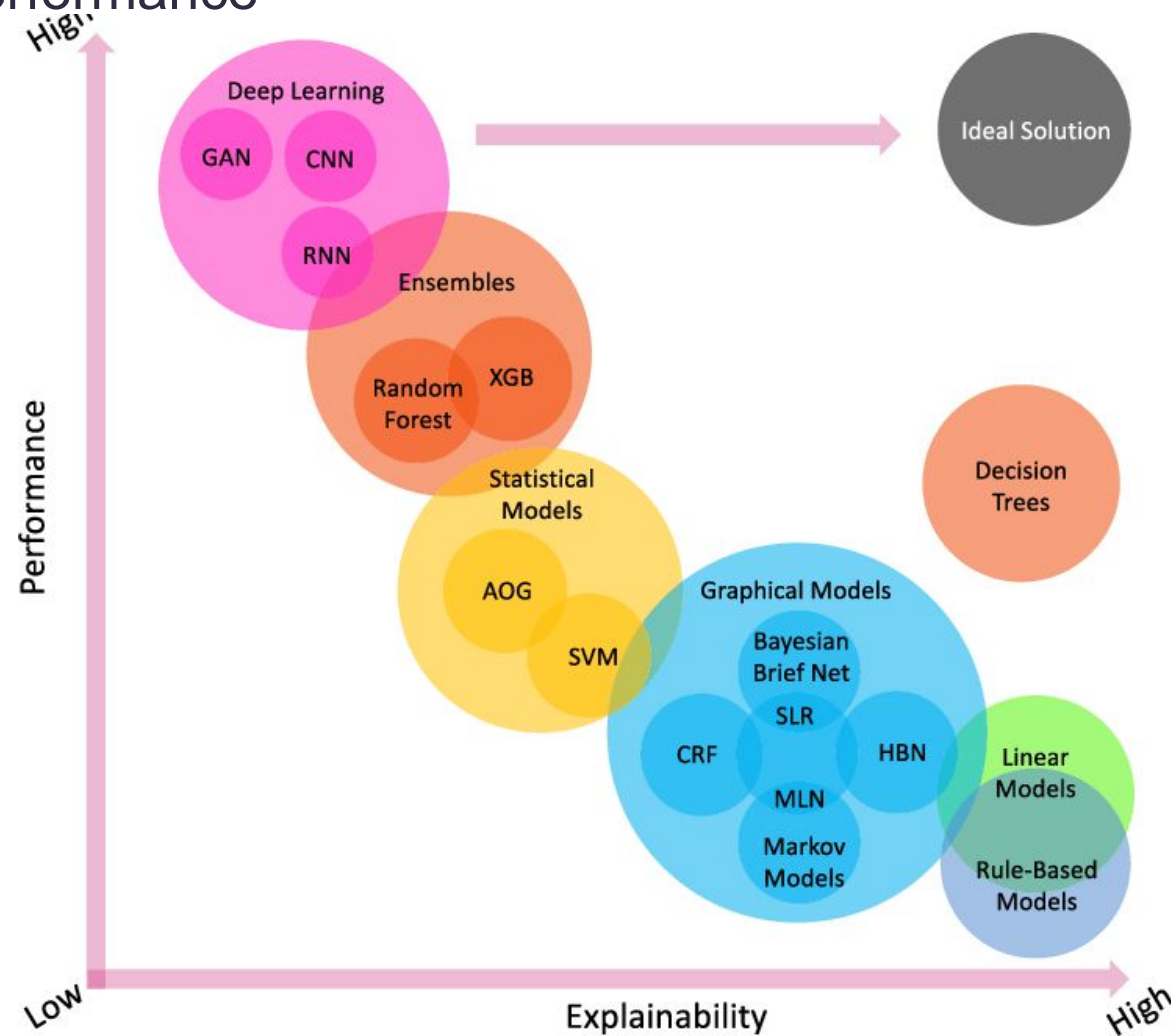
NN Learning Rate

The Goldilocks Problem of Parameter Tuning



NotebookLM

Explainability vs. Performance



https://www.researchgate.net/figure/Model-explainability-vs-model-performance-for-widely-used-machine-learning-and-deep_fig1_353621660

NN -> Transformers

Transformers Replace CNNs, RNNs

Transformers are in many cases replacing convolutional and recurrent neural networks (CNNs and RNNs), the most popular types of deep learning models just five years ago.

Indeed, 70 percent of [arXiv](#) papers on AI posted in the last two years mention transformers. That's a radical shift from a [2017 IEEE study](#) that reported RNNs and CNNs were the most popular models for pattern recognition.

No Labels, More Performance

Before transformers arrived, users had to train neural networks with large, labeled datasets that were costly and time-consuming to produce. By finding patterns between elements mathematically, transformers eliminate that need, making available the trillions of images and petabytes of text data on the web and in corporate databases.

In addition, the math that transformers use lends itself to parallel processing, so these models can run fast.

Transformers now dominate popular performance leaderboards like [SuperGLUE](#), a benchmark developed in 2019 for language-processing systems.

<https://blogs.nvidia.com/blog/what-is-a-transformer-model/#:~:text=A%20transformer%20model%20is%20a,the%20words%20in%20this%20sentence.&text=If%20you%20want%20to%20ride,in%20AI%2C%20grab%20a%20transformer.>



Q&A

Context Optimization

What is told to the model

